Dynamic Programming Volumes I and II

A Quick Introduction — Lecture 1 of 2

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These Lectures

An introduction to DP1 and DP2: a general framework for dynamic programming

https://dp.quantecon.org

1. Lecture 1: Background and motivation
2. Lecture 2: Theory and applications

A fast-paced introduction for an advanced audience

Flow

Lecture 1: Motivation

1. Review of some familiar DP problems — and the standard theory
2. Variations on a theme — new perspectives, new problems

Lecture 2: Theory

3. Introduction to the abstract theory — one theory to rule them all (okay, not all)
4. Applications

What's DP?

Dynamic programming is a recursive technique for solving optimization problems

Vast range of applications across econ / finance / OR / CS / etc.

- optimal savings
- real options
- firm investment
- retirement planning
- portfolio allocation
- monetary policy

- fiscal policy
- job search
- dynamic pricing
- capacity allocation
- social media feeds
- career choice
- economic geograph
- production chains
- etc., etc.

- natural language processing
- sequential analysis
- automotive cruise control
- hybrid vehicle power management
- LaTeX typesetting
- battery consumption
- compiler optimizations
- etc., etc.

- protein structure prediction
- RNA folding
- Optimal control of chemical reactors
- peptide sequencing
- phylogenetic reconstruction
- chemical isomer generation
- etc., etc.

etc., etc., etc.

The foundation stone for reinforcement learning (RL)

- Q-learning
- Actor-critic
- etc.

Eg. LLM post-training

> "DeepSeek-R1 researchers showed that RL can induce emergent reasoning behaviours: self-verification, backtracking, exploring alternative solution paths. The model wasn't taught these strategies — they emerged from the RL optimization pressure."
>
> — Claude

Fun fact: Economists' assumption that firms use DP is getting more accurate

- Dynamic pricing
- Capacity allocation
- Social media feeds
- Supply chain optimization
- etc.
- etc.

Modern Trends: Variations on a Theme

- recursive preferences
- ambiguity aversion
- adversarial agents / robust control
- nonlinear discounting
- nonstandard representations (e.g., Q-factors, post-action value functions)
- distributional RL
- etc., etc. ...

No general theory that covers all these cases

DP1–DP2: Value Added

What's in these books?

- A new abstract DP framework — inspired by and builds on Bertsekas (2022)
  - Built to handle the applications described above
- Some new results
  - isomorphisms, factorizations, etc.
- A large range of applications

Abstract DP framework?

- An **abstract dynamic program** is a family of maps $\mathbb{T} = \{T_\sigma : \sigma \in \Sigma\}$ on a partially ordered set...
- The **Bellman equation** is $v = \bigvee_{\sigma \in \Sigma} T_\sigma v$ where $\bigvee_{\sigma \in \Sigma}$ is the supremum
- etc., etc. — explanations in Lecture 2

Now add various theorems:

- The value function satisfies the Bellman equation if and only if ...

Motivation for the Framework

Motivation 1 Generality

- Captures a large range of DP applications / variations
- One theory for all, not one theory per problem
- New results (general optimality results, factorizations, etc.)

Motivation 2 Simplicity

- Proofs of key ideas are "elementary"
- Maximizes effectiveness of machine reasoning

Machine Reasoning

AI-friendly logical framework:

> "1. The reasoning is algebraic, not geometric or topological. This is the kind of thing transformer architectures handle well – pattern matching over structured symbolic expressions.
>
> 2. The definitions do the heavy lifting. This is a sign of good mathematical design and also makes the proofs easy for AI: there aren't many "creative" steps, just systematic application of the definitions.
>
> 3. Your work states every hypothesis, names every property, and never hand-waves. This is exactly the input format AI works best with."

— Claude

Application: Inventory Management

We begin by reviewing an inventory problem

- Can be handled by well-known theory

Motivation:

- Standardize terminology and notation
- Recall key optimality results
- Recall how the proofs work

Inventory dynamics:

$$X_{t+1} = \max(X_t - D_{t+1}, 0) + A_t$$

inventory

demand

order of new stock

Profits:

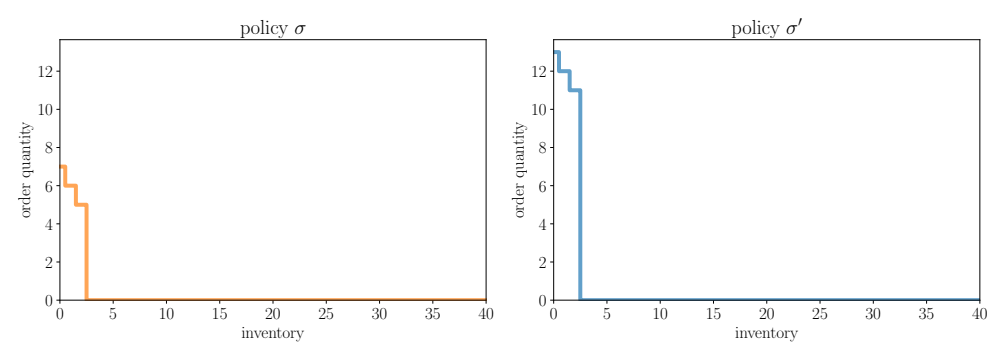
$$\pi_{t+1} = \min(X_t, D_{t+1}) - c\,A_t - \kappa \mathbb{1}\{A_t > 0\}$$

revenue

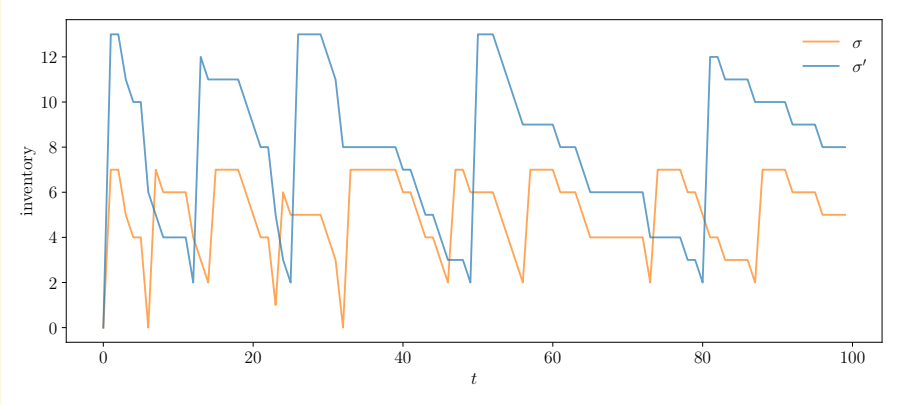
unit cost

fixed cost

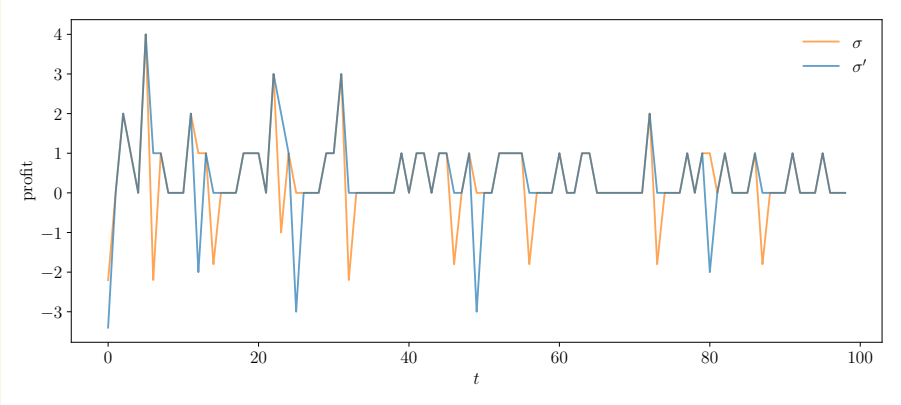
A policy σ maps current inventory to current order size



Inventory dynamics $(X_t^{\sigma})_{t\geqslant 0}$ and $(X_t^{\sigma'})_{t\geqslant 0}$ under σ and σ'



Profit sequences $(\pi_t^{\sigma})_{t\geqslant 0}$ and $(\pi_t^{\sigma'})_{t\geqslant 0}$ under σ and σ'



Profits — present value

$$\Pi_\sigma(x) = \sum_{t \geqslant 0} \beta^t \pi^\sigma_{t+1}$$

Depends on

1. policy $\sigma \in \Sigma :=$ set of all policies
2. initial inventory x
3. luck: demand sequence $D_1, D_2, \ldots$

EPV:

$$v_\sigma(x) := \mathbb{E}[\Pi_\sigma(x)] = \mathbb{E}\left[\sum_{t \geqslant 0} \beta^t \pi^\sigma_{t+1}\right]$$

EPV recursion:

expected present value under σ given current state

$$v_\sigma(x) = \mathbb{E}[\pi_\sigma(x, D)] + \beta \mathbb{E}[v_\sigma(h_\sigma(x, D))]$$

current profits under σ

next period inventory under σ

Equivalently, v_σ is the fixed point of the **policy operator**

$$(T_\sigma\, v)(x) = \mathbb{E}\,[\pi_\sigma(x, D)] + \beta \mathbb{E}\,[v(h_\sigma(x, D))]$$

Repeating, the policy operator is

$$(T_\sigma\, v)(x) = \mathbb{E}\left[\pi_\sigma(x, D)\right] + \beta \mathbb{E}\left[v(h_\sigma(x, D))\right]$$

Properties

- A contraction map with unique fixed point $v_\sigma \in V := \mathbb{R}^{\mathsf{X}}$
- Order-preserving: $v \leqslant w \implies T_\sigma v \leqslant T_\sigma w$

Computation: Fix v and iterate to get $\lim_{k\to\infty} T_\sigma^k v = v_\sigma$

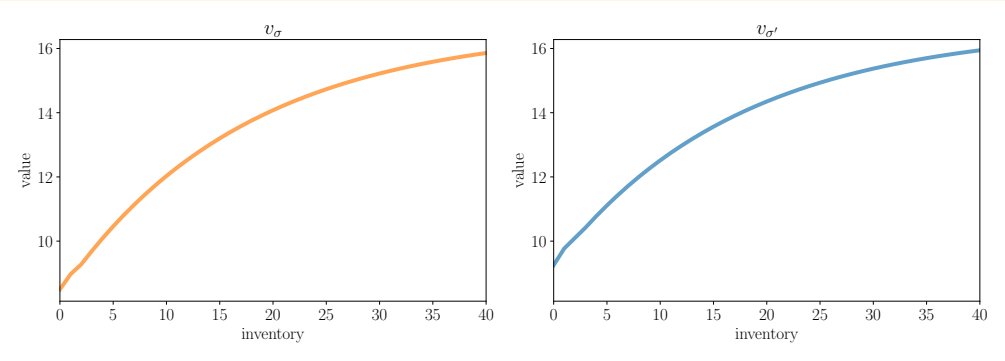


Figure 1: Lifetime value functions v_σ and $v_{\sigma'}$

Value function:

$$v_\Delta(x) := \sup_{\sigma \in \Sigma} v_\sigma(x)$$

A policy $\sigma \in \Sigma$ is called **optimal** if $v_\sigma = v_\Delta$

Equivalent:

$$v_\sigma(x) \geqslant v_\tau(x) \text{ for every } \tau \in \Sigma \text{ and every } x \in \mathrm{X}$$

- Does such a policy exist??
- If so, how should we compute it?

The **Bellman operator** is

$$(T_\Delta v)(x) = \max_{a \in \Gamma(x)} \mathbb{E}[\pi(x, a, D)] + \beta \mathbb{E}[v(h(x, a, D))]]$$

Function v satisfies the **Bellman equation** if $T_\Delta v = v$

Given any v, a policy σ is called **v-greedy** if

$$\sigma(x) \in \arg\max_{a \in \Gamma(x)} \mathbb{E}[\pi(x, a, D)] + \beta \mathbb{E}[v(h(x, a, D))] \qquad \forall\, x \in \mathsf{X}$$

Note: This is equivalent to $T_\sigma v = T_\Delta v$

Theorem 1: Optimality

The following statements are true

1. The value function v_Δ is the unique solution to the Bellman equation in V
2. A policy σ is optimal if and only if it is v_Δ-greedy
3. At least one optimal policy exists

Below we will say that the "fundamental optimality properties hold"

Proof

We skip details but note that the following facts are used:

- Each T_σ is order preserving ($v \leqslant v'$ implies $T_\sigma\, v \leqslant T_\sigma\, v'$)
- Each T_σ is a contraction map
- $T_\triangle$ is order preserving ($v \leqslant v'$ implies $T_\triangle\, v \leqslant T_\triangle\, v'$)
- $T_\triangle$ is a contraction map

Proof due to Bellman, Blackwell — see, e.g., Stokey and Lucas (1989) or Puterman (2005)

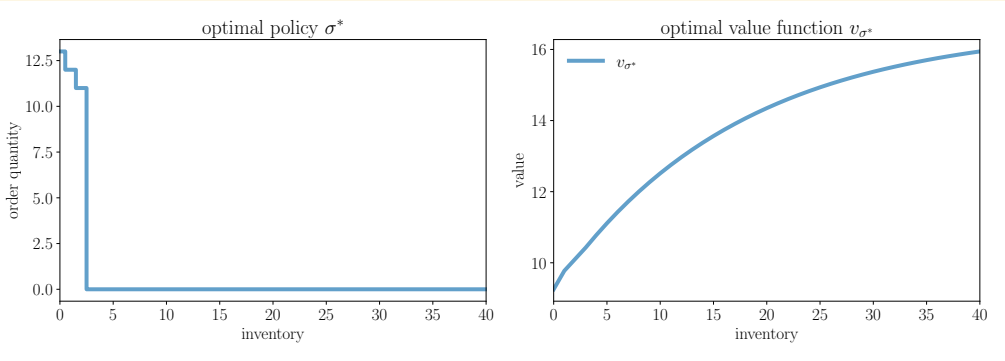


Figure 2: Optimal policy and value function

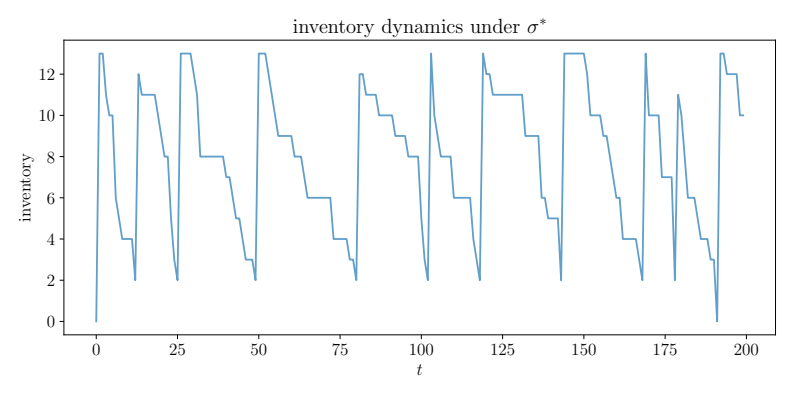


Figure 3: Inventory dynamics under the optimal policy

Algorithms

The big three:

1. Value function iteration (VFI)
2. Howard policy iteration (HPI)
3. Optimistic policy iteration (OPI)

There are countless variations on these ideas

- Eg. actor-critic, core method for RL, is a variation on OPI

Algorithm 1: Value function iteration (VFI)

input $v \in V$, an initial guess of v_Δ

input `tol`, a tolerance level for error

$\varepsilon \leftarrow \texttt{tol} + 1$

while $\varepsilon > \texttt{tol}$ **do**

$\quad v' \leftarrow T_\Delta v$

$\quad \varepsilon \leftarrow \|v' - v\|$

$\quad v \leftarrow v'$

end

return a v-greedy policy σ

Algorithm 2: Howard policy iteration (HPI)

input $\sigma \in \Sigma$ and tolerance `tol`

set $v \leftarrow$ the lifetime value v_σ

set $\varepsilon \leftarrow$ `tol` $+ 1$

while $\varepsilon >$ `tol` **do**

  $\sigma \leftarrow$ a v-greedy policy

  $v' \leftarrow$ the lifetime value v_σ

  $\varepsilon \leftarrow \|v' - v\|$

  $v \leftarrow v'$

end

return σ

Algorithm 3: Optimistic policy iteration (OPI)

input $m \in \mathbb{N}$ and tolerance $\texttt{tol}$

input $v \in V$, an initial guess of v_Δ

$\varepsilon \leftarrow \texttt{tol} + 1$

while $\varepsilon > \texttt{tol}$ **do**

  $\sigma \leftarrow$ a v-greedy policy

  $v' \leftarrow T_\sigma^m v$

  $\varepsilon \leftarrow \|v' - v\|$

  $v \leftarrow v'$

end

return σ

Loosely speaking

- VFI is fixed point iteration on the Bellman operator
- HPI is Newton iteration on the Bellman operator
- OPI is a convex combination of the two

For the inventory problem we have the following results:

Theorem 2: Convergence of Algorithms

The following statements are true

1. The sequence of iterates $(v_k^{\mathrm{VFI}})_{k\geqslant 0}$ generated by VFI converges to v_Δ
2. The sequence of iterates $(v_k^{\mathrm{OPI}})_{k\geqslant 0}$ generated by OPI converges to v_Δ
3. The sequence of iterates $(v_k^{\mathrm{HPI}})_{k\geqslant 0}$ generated by HPI converges to v_Δ

In addition, HPI returns an exact optimal policy in finitely many steps

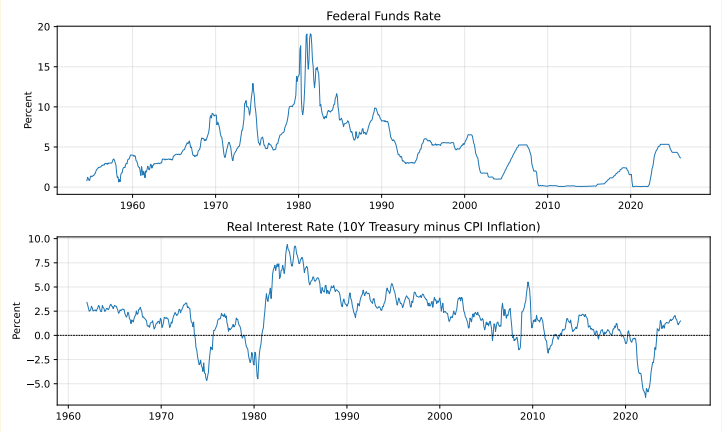
Extensions

Until now we have

- studied a traditional problem
- recalled standard results
- reviewed how the proofs work (operators are order preserving contractions)
- discussed the three major algorithms

Now let's look at some variations

Extension 1: State-Dependent Discounting



Now β depends on an exogenous state

The new Bellman equation is

$$v(x, z) = \max_{a \in \Gamma(x)} \mathbb{E} \left[\pi(x, a, D) + \beta(z) \int v(h(x, a, D), z') \, Q(z, \mathrm{d}z') \right]$$

Theoretical issues

- operators T_σ, T_Δ are still order-preserving (when $\beta \geqslant 0$)
- but contraction properties can fail

Do the fundamental optimality properties hold? Do the algorithms converge?

Extension 2: Handling Risk

"When we tested how 1,500 executives from 90 countries reacted to different investment scenarios, we discovered that they demonstrated extreme levels of risk aversion regardless of the size of the investment."

— Koller et al., Overcoming a Bias Against Risk (McKinsey, 2017)

"Managers acknowledged that although their risk aversion was bad for their companies, it was good for their careers."

— Rudd & Lawson, Your Company Is Too Risk-Averse (HBR, 2020)

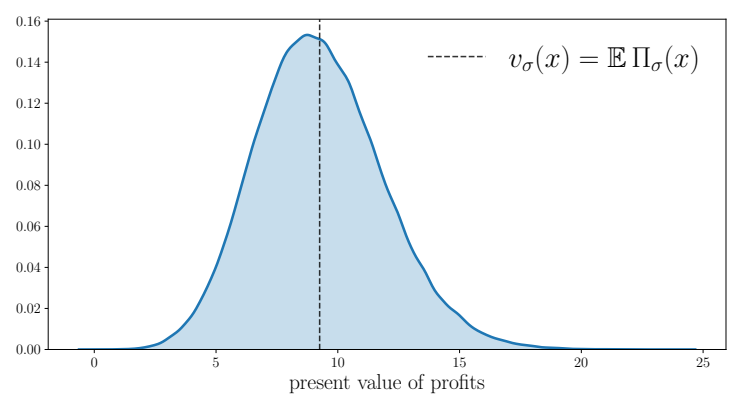
How to Handle Risk Preferences

Recall that

$$\Pi_\sigma(x) := \sum_{t \geqslant 0} \beta^t \pi_t^\sigma = \text{ present value of profits under } \sigma$$

By definition we have

$$\text{EPV } = v_\sigma(x) = \mathbb{E}\left[\Pi_\sigma(x)\right]$$



Distribution of $\Pi_\sigma(x) = \sum_{t \geqslant 0} \beta^t \pi_t^\sigma$

Optimal policy σ^* solves

$$v_{\sigma^*}(x) = \max_{\sigma \in \Sigma} \; v_\sigma(x) = \max_{\sigma \in \Sigma} \; \mathbb{E}\left[\Pi_\sigma(x)\right]$$

But what if we care about risk?

Then we should solve

$$\max_{\sigma \in \Sigma} \; \mathscr{E}\left[\Pi_\sigma(x)\right] \quad \text{for suitable nonlinear } \mathscr{E}$$

Eg. Mean-variance criterion

$$\mathscr{E}[\Pi_\sigma(x)] = \mathbb{E}[\Pi_\sigma(x)] - \frac{\gamma}{2}\,\mathrm{Var}[\Pi_\sigma(x)]$$

Eg. Mean minus value-at-risk

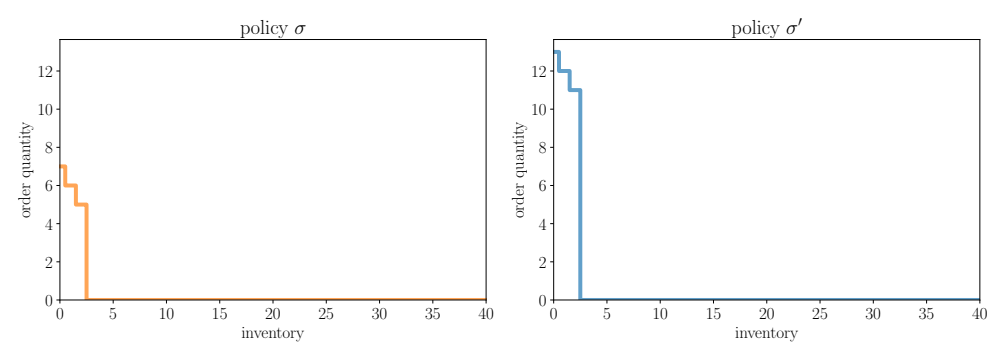
$$\mathscr{E}[\Pi_\sigma(x)] = \mathbb{E}[\Pi_\sigma(x)] - \frac{\gamma}{2}\,\mathrm{VaR}_\alpha[\Pi_\sigma(x)]$$

Eg. Kreps–Porteus expectation

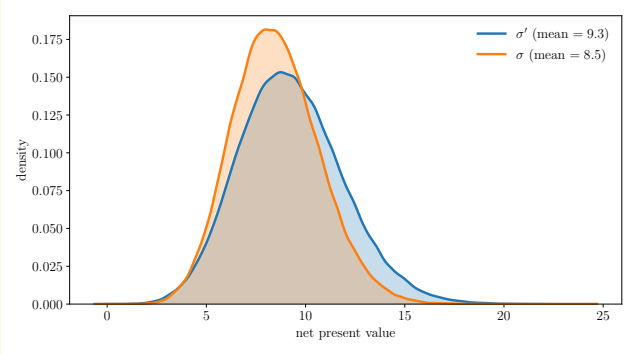
$$\mathscr{E}[\Pi_\sigma(x)] = \left\{\mathbb{E}[\Pi_\sigma(x)^{1-\gamma}]\right\}^{1/(1-\gamma)}$$

Now which policy is best?

Recall these two policies:

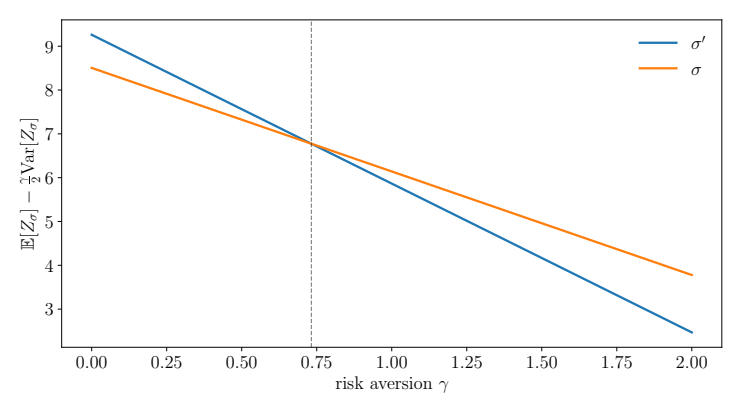


Distributions of $\Pi_\sigma(x)$ and $\Pi_{\sigma'}(x)$ under σ and σ'



Policy σ' has higher mean but more spread

Under mean-variance, for large γ, the conservative policy σ is preferred:



Can we **use DP theory** to solve $\max_{\sigma \in \Sigma} \mathscr{E}[\Pi_\sigma(x)]$ for nonlinear $\mathscr{E}$??

- In general we can't!
- Even the question is wrong: nonstationary policies can do better
- And this leads to time inconsistency...

Researchers are trying to fix these problems with varying degrees of success

- A nice example is Bäuerle and Jaśkiewicz (2025)

Back to Recursion

Workaround: Instead of $\mathscr{E}[\Pi_\sigma(x)]$, apply $\mathscr{E}$ **period by period**

Eg. Lifetime value of a policy is a fixed point v_σ of

$$(T_\sigma v)(x) = \mathscr{E}\left[\pi_\sigma(x, D) + \beta v(h_\sigma(x, D))\right]$$

Here

- $\mathscr{E}$ is a possibly nonlinear "risk-adjusted expectation"
- T_σ is a new **policy operator**

Valuations are recursive **by construction**

Recursive...but traditional theory does not cover this case

Do the fundamental optimality properties hold?

Do the big three algorithms converge?

- T_σ and T_Δ are order preserving when $Z \leqslant Z' \implies \mathscr{E}(Z) \leqslant \mathscr{E}(Z')$
- But not always contractions

Does a modified proof still go through?

Some Additional DP Problems

We want a general theory of DP that can cover the preceding cases

But wait!

There are still more interesting DP problems

Let's list some more DP problems to get a more complete accounting of what we need

Example: Sequential Analysis (Wald)

Hidden state is θ_0 or θ_1 and π = current belief that $\theta = \theta_1$

Actions:

- stop and choose $\theta = \theta_0$
- stop and choose $\theta = \theta_1$
- draw another observation at cost c

The Bellman equation:

$$g(\pi) = \min \left\{ \pi L_0,\ (1-\pi) L_1,\ c + \int g(\pi')\, P(\pi, \mathrm{d}\pi') \right\}$$

Challenging because no discounting — T_σ and $T_\triangledown$ are order preserving but not contractions

Example: Stochastic Shortest Path

An agent moves through a finite graph, choosing successor nodes to reach a target node 0 at minimum expected cost

The Bellman equation is

$$v(x) = \min_{a \in \Gamma(x)} \left\{ c(x, a) + \sum_{x'} v(x') P(x, a, x') \right\} \qquad (x \neq 0), \qquad v(0) = 0$$

- $c(x, a) > 0$ is the cost of choosing edge a at node x
- $P(x, a, x')$ is the transition probability
- No discounting — T_σ and $T_\triangledown$ are order preserving but not contractions

Example: Epstein–Zin

A popular specification in finance / macro-finance

The Bellman equation is

$$v(w) = \max_{0 \leqslant c \leqslant w} \left[(1-\beta) c^{1-1/\psi} + \beta \left(\int v(R(w-c)+y)^{1-\gamma} \varphi(\mathrm{d}y) \right)^{\frac{1-1/\psi}{1-\gamma}} \right]^{\frac{1}{1-1/\psi}}$$

- T_σ and T_Δ are order preserving — but not contracting in general
- Do the fundamental optimality properties hold?
- How about convergence of algorithms?

Example: irreversible Investment

An irreversible investment problem: the firm chooses investment $i \geqslant 0$ to maximize present value of profits

$$v(k, z) = \max_i \left\{ f(k, z) - i + \beta \int v(k', g(z, \xi)) \varphi(\mathrm{d}\xi) \right\}$$

- $k' := i + (1 - \delta)k$
- max is over i with $0 \leqslant i \leqslant \theta f(k, z)$

This is a standard DP problem

However...

Robust Control

Suppose now that the manager does not fully trust the reference model

She fears misspecification in terms of the distribution φ of the shock sequence

She chooses to maximize against worst-case alternatives

$$v(k,z) = \max_{0 \leqslant i \leqslant \theta f(k,z)} \left\{ f(k,z) - i + \beta \inf_{\psi \ll \varphi} \left[\int v(k', g(z,x)) \psi(\mathrm{d}x) + \frac{1}{\gamma} D(\psi \,\|\, \varphi) \right] \right\}$$

- Do the fundamental optimality properties hold?
- Do the big three algorithms converge?

Example: Nonlinear Discounting

Job search: a worker receives wage offers (W_t) and chooses when to accept

$$v(w) = \max\left\{\frac{w}{1-\beta},\ c + \beta \int v(w')\varphi(\mathrm{d}w')\right\}$$

With nonlinear discounting, β depends on the continuation value:

$$v(w) = \max\left\{e(w),\ c + \int \beta[v(w')]P(w, \mathrm{d}w')\right\}$$

- Example: can model "magnitude effects" — large rewards discounted less
- T_σ, $T_\triangle$ are order preserving but not generally contracting

Negative Discount Rates

Consider a DP problem with Bellman equation

$$v(x) = \min_{0 \leqslant a \leqslant x} \{c(a) + \delta\, v(x - a)\}$$

- c measures cost of effort with $c' > 0$ and $c'' > 0$
- $\delta > 1$ for negative discount rates — want to get task "over and done with"

Properties:

- The Bellman operator is order-preserving but not a contraction — since $\delta > 1$
- Do the fundamental optimality properties hold? Do the algos converge?

There is a connection to production chains:

Consider a firm that purchases at stage t, sells at s, in-house costs $= c(s-t)$

With transaction cost $\delta > 1$, profits are

$$\pi = p(s) - c(s-t) - \delta\, p(t)$$

Minimizing costs and setting profits to zero, equilibrium prices obey

$$p(s) = \min_{0 \leqslant t \leqslant s} \{c(s-t) + \delta\, p(t)\}$$

Note that p is an equilibrium price function $\iff$ p solves the Bellman equation of the negative discount problem

Example: Structural Estimation

Model agents as solving DPs; infer parameters θ to match model outputs to data

Post-action value function g satisfies a nonstandard Bellman equation:

$$g(x, a) = \int \max_{a' \in \mathsf{A}} \left[r(x', a') + \beta g(x', a') \right] P(x, a, \mathrm{d}x')$$

- Integration and maximization are **reversed** relative to standard DP
- g depends on both state x and action a
- Doesn't fit classical DP framework or Bertsekas (2022) — how can we handle it?

Example: Distributional DP/RL

Setup: Finite MDP with Bellman equation

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \sum_{x'} v(x') P(x, a, x') \right\}$$

Traditional DP maximizes the expected discounted return $v_\sigma(x) = \mathbb{E}\left[Z_\sigma(x)\right]$ where

$$Z_\sigma(x) = \sum_{t \geqslant 0} \beta^t r(X_t, \sigma(X_t))$$

Here

- $X_0 = x$
- $X_{t+1} \sim P(X_t, \sigma(X_t), \cdot)$

Distributional DP tracks an estimate of the distribution of $Z_\sigma(x)$

- Captures variance, quantiles, tail risk
- Core idea behind distributional RL (Bellemare et al. 2017)

Key object: A **distributional value function** is a map $x \mapsto \eta(x, \cdot)$ assigning a return distribution to each state

- Traditional: $v(x)$ represents expected present value from state x
- Distributional: $\eta(x, \cdot)$ represents distribution of present values from state x

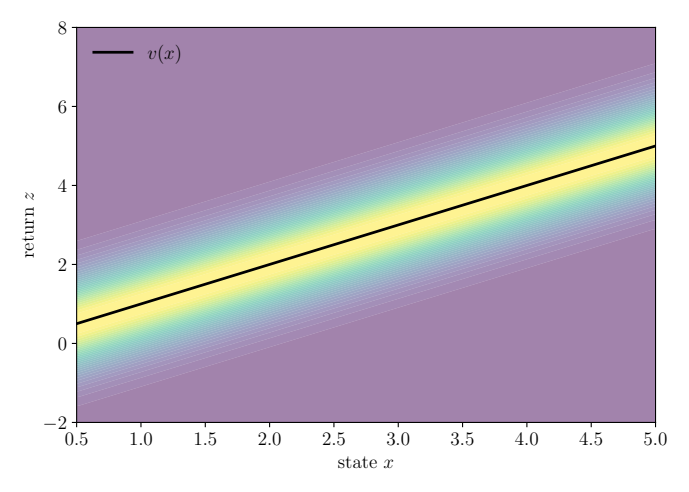


Figure 4: Distributional value function $\eta(x, \cdot)$ and scalar value $v(x)$

The **distributional policy operator** D_σ maps distributional value functions to distributional value functions:

$$(D_\sigma \eta)(x, B) = \int \left[\int \mathbb{1}_B(r_\sigma(x) + \beta v)\, \eta(x', \mathrm{d}v) \right] P_\sigma(x, \mathrm{d}x')$$

Key properties:

- D_σ is **order preserving** under pointwise stochastic dominance
- D_σ is a β-**contraction** in the supremum Wasserstein-1 metric
- The unique fixed point $\eta_\sigma(x, \cdot)$ gives the distribution of $Z_\sigma(x)$ at each x

The scalar value function is recovered: $v_\sigma(x) = \int z\, \eta_\sigma(x, \mathrm{d}z)$

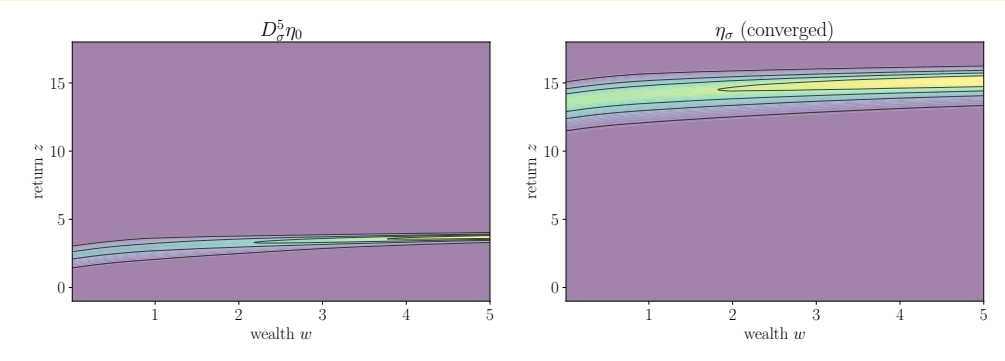


Figure 5: Iterating D_σ: early iterate and converged η_σ

Questions:

- Can we build an abstract theory general enough to handle both distributional and traditional DP?
- If so, what can we learn about distributional DP?

State of Play

The set of possible applications is extremely broad

Traditional assumptions often fail:

- failure of contractivity
- failure of additive separability
- nonstandard timing
- nonstandard value spaces

and some are preserved

- recursivity, order-preserving T_Δ and T_σ

DP1–DP2 but an abstract DP framework that can handle these cases

- DP1 considers the case of finite states and actions — cleanest
- DP2 considers the general case

Lecture 2 gives an introduction to the key ideas

References

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